

HOJAE SON

Amherst, MA · hojaeson@umass.edu

RESEARCH INTEREST

Systems for Machine Learning, Vector Database, Retrieval Augmented Generation (RAG), Graph Neural Network (GNN), Transformer Inference

EDUCATION

University of Massachusetts, Amherst

Ph.D., Computer Sciences *GPA: 3.68/4.0*

Amherst, MA

Sep 2023 - Present

Kyungpook National University

MS., Computer Sciences *GPA: 4.18/4.50*

South Korea

March 2017 - Feb 2019

Kyungpook National University

BS., Computer Sciences *GPA: 4.11/4.50*

South Korea

March 2010 - Feb 2017

RESEARCH PROJECTS

Transformer Inference/Fintuning

- Semantic-Operator Query Optimization with Transformer Inference in vLLM
- Optimizing Language Model with KV Cache in vLLM/HuggingFace Transformer
- Efficient fine-tuning of language models for RAG pipelines

Graph Neural Network (GNN) Training

- Full-batch and Mini-Batch comparison on GNN Training

PUBLICATIONS

- [1] Md Ashraful Islam, **Hojae Son**, Suhaas Kiran, and Marco Serafini. "PRISM: A Training System to Unlock the Potential of Temporal Graph Learning Through Staleness Avoidance." In PVLDB 2026.
- [2] Saurabh Bajaj, **Hojae Son**, Juelin Liu, Hui Guan, and Marco Serafini. "Graph Neural Network Training Systems: A Performance Comparison of Full-Graph and Mini-Batch." In VLDB 2025.
- [3] **Hojae Son**, Anand Paul, Gwanggil Jeon. "Country Interest Analysis Based on Long-Term Short-Term Memory" In ICDM Workshops 2018.
- [4] M. Mazhar Rathore, **Hojae Son**, Awais Ahmad, Anand Paul. "Real-time video processing for traffic control in smart city using Hadoop ecosystem with GPUs" In Soft Computing 2017.
- [5] M. Mazhar Rathore, **Hojae Son**, Awais Ahmad, Anand Paul, Gwanggil Jeon. "Real-Time Big Data Stream Processing Using GPU with Spark over Hadoop Ecosystem" In Parallel Programming 2017.

WORK EXPERIENCE

Unity Cluster

Research Assistant

Amherst, MA

Jan 2024 to Present

- Assist researchers with Unity HPC platform (Slurm) usage, including workflow execution and AI dataset management
- Distributed System training, Inference settings for machine learning researcher

LuluLab, Inc. (Samsung C-Lab)

Research Engineer

South Korea

Oct 2021 - Dec 2022

- Develop SDK for AI Skin Analyzer; Integrate 7 deep learning models into a system
- Build RESTful service with AWS ECS pipeline

Hyundai Mobis, Inc.

Research Engineer

South Korea

Jan 2019 - Feb 2021

- Develop Semantic Segmentation models for ADAS VP project
- Collaborate with the Texas Instrument ADAS sector to tackle embedding deep learning on Genesis 90, using the TIDL library / ONNX

PATENT

Hojae Son, Jaeyoung Lee. "System for estimating shape of road based on deep learning network and method thereof" (Korean Patent number: 10-2020-0059386) Hyundai Mobis, 2020

TECHNICAL SKILLS

C/C++, Python, Pytorch, CUDA, Git, DVC, Linux, Deep learning, Machine learning, Java, Android SDK/NDK

TEACHING ASSISTANT

- Database Fall, 2023
- Artificial Intelligence Summer, 2017
- Java Programming Spring, 2016
- Python Programming Fall, 2016
- Java Programming Spring, 2016
- C++ Programming Fall, 2015